Python Basics - Assignment 1 TODO(1)

September 29, 2021

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1 Python Basics - Assignment 1 TODO

Task 1: Create a string (name), an integer (num) and a floating point number (fnum). Assign your name to name, 25 to num and 56.78 to fnum. Print the output for all 3 variables.

```
###name - use your own name
#num
#fnum
#print name
#print num
#print fnum
##continue to code solution cell below.
```

[1]:
```
name = "Faruk"
num = 25
fnum = 56.78
print (name)
print (num)
print (fnum)
```

```
Faruk
25
56.78
```

1.0.1 Example solutions task 1.

CLICK ME

```
name = "Eero"
num = 25
fnum = 56.78
print (name)
print (num)
print (fnum)
```

Task 2: Use the capitalize() method to convert the first character of a string to uppercase letter. Continue to code solution cell below.

```
mystring = "this is getting interesting."
mycapital_string = #use capitalize method
print the capitalized string
```

[3]:
```
mystring = "this is getting interesting."
mycapital_string = mystring.capitalize()
print (mycapital_string)
```

```
This is getting interesting.
```

Example solutions task 2. CLICK ME

```
# Example
mycapital_string = mystring.capitalize()
print (mycapital_string)
```

Task 3: The find() method returns the index of first occurence of the substring if it is found and if it does not find it returns -1. Use find() method to complete the code below.

```
text = "Python is an easy to learn language"
result = #find learn
print (result)
```

[5]:
```
text = "Python is an easy to learn language"
result = text.find("learn")
print (result)
```

```
21
```

Example solutions task 3 CLICK ME

```
# Example
result = text.find('learn')
print (result)
```

Task 4: It is possible to assign the same value to multiple variables at once in Python. Complete the code below.

```
# create 3 variables a,b & c
# assign 1 as value to all variables in the same statement
# incorrect answer if it is
a=1 b=1 c=1

#print a
#print b
#print c
```

[9]:
```
a = b = c = 1
print (a)
print (b)
print (c)
```

```
1
1
1
```

Example solutions task 4 CLICK ME

```
# Example
a = b = c = 1
print (a)
print (b)
print (c)
```

Task 5: Use for and if to find and print all names from list that are at least 5 characters long. `names = ["Alice", "Bob", "Charlie", "Dave", "Emily"] for # Loop through the list of names # Print all names that are 5 characters or longer # Use len method to find length of the string`

[10]:
```
names = ["Alice", "Bob", "Charlie", "Dave", "Emily"]
for x in names:
    if len(x) >= 5:
        print(x)
```

```
Alice
Charlie
Emily
```

Example solutions task 5 CLICK ME

```
# Example
names = ["Alice", "Bob", "Charlie", "Dave", "Emily"]
for n in names:
    if len(n) >= 5:
        print (n)
```

Task 6: Create a list named mylist and include 5 different data items.

```
mylist =   # it should contain 5 different data items, e.g. mylist = ['Apple', 2, 3.1415, 'd',
print() # print the entire list
print() # print only the second element, note that the first element index is 0
#use append to add one more item to your list
print() #print the entire list
```

[15]:
```
mylist = ['Apple', 2, 3.1415, 'd', 1024 ]
print(mylist)
print(mylist[1])
mylist.append("faruk")
print(mylist)
```

```
['Apple', 2, 3.1415, 'd', 1024]
2
```

```
['Apple', 2, 3.1415, 'd', 1024, 'faruk']
```

Example solutions task 6 CLICK ME

```
mylist = ['Apple', 2, 3.1415, 'd', 1024 ]
print (mylist)
print (mylist[1])
mylist.append("new_item")
print (mylist)
```

Task 7: Tuple

```
courses = ("Introduction to Basics for Embedded Python", Python Basics for IoT, "Python Project
print() #use len method to print the number of items(courses) in the tuple courses.
print() #print all items in the tuple courses
print() #print the third item in the tuple
```

[19]:
```
courses = ("Introduction to Basics for Embedded Python", "Python Basics for␣
↪IoT", "Python Project")
print (len(courses))
print (courses)
print (courses[2])
```

```
3
('Introduction to Basics for Embedded Python', 'Python Basics for IoT', 'Python
Project')
Python Project
```

Example solutions task 7 CLICK ME

```
courses = ("Introduction to Basics for Embedded Python", "Python Basics for IoT", "Python Proje
print (len(courses))
print (courses)
print (courses[2])
```

Task 8: Dictionary

```
#create a dictionary named test with the keys 'course', 'time' and 'ects' and set values as 'Py
test = {

}
print() #use get method to print the value of the course
test[] =  #change the course value to Python Basics for IoT

test[] = #add the key/value pair as year:2020 to the test dictionary
print() #print the entire dictionary
#use pop method to remove time form the dictionary
print() #print the dictionary
```

[23]:
```
test = {
    'course': 'python basic',
    'time': 'autumn',
    'ects': '5'
}

print (test.get('course'))
test['course'] = 'Python Basics for IoT'
test['year'] = 2020
test.pop('time')
print(test)
```

```
python basic
{'course': 'Python Basics for IoT', 'ects': '5', 'year': 2020}
```

Example solutions task 8 CLICK ME

```
test = {
    'course': 'Python Basics',
    'time': 'Autumn',
    'ects': 5
}
print (test['course'])
test['course'] = 'Python Basics for IoT'
test['year'] = 2020
print (test)
test.pop('time')
print (test)
```

Task 9: Create Integer, String and Floating point number

The target of this exercise is to create a string, an integer, and a floating point number. The string should be named mystring and should contain the word "hello". The floating point number should be named myfloat and should contain the number 10.0, and the integer should be named myint and should contain the number 20. Test your variables using print so that program prints variable only if it contains specifig value.Note that you have to use indentation between if-clauses. Python uses indentation for blocks, instead of curly braces. Both tabs and spaces are supported, but the standard indentation requires standard Python code to use four spaces. For example:

```
x = 1
if x == 1:
    # indented four spaces
    print("x is 1.")
```

Test code like that way:

```
if mystring == "hello":
    print("String: %s" % mystring)
```

[26]:
```
mystring = "hello"
myfloat = 10.0
myint =20

if mystring == "hello":
    print("String: %s" % mystring)
if isinstance(myfloat, float) and myfloat == 10.0:
    print("Float: %f" % myfloat)
if isinstance(myint, int) and myint == 20:
    print("Integer: %d" % myint)
```

```
String: hello
Float: 10.000000
Integer: 20
```

Example solutions task 9 CLICK ME

```
mystring = "hello"
myfloat = 10.0
myint = 20

# testing code
if mystring == "hello":
    print("String: %s" % mystring)
if isinstance(myfloat, float) and myfloat == 10.0:
    print("Float: %f" % myfloat)
if isinstance(myint, int) and myint == 20:
    print("Integer: %d" % myint)
```

Task 10: Convert Celsius to Fahrenheit

The target of this exercise is to code Python Program which take the temperature in Celsius and convert it to Fahrenheit.

Problem Solution

1. Take the value of temperature in Celsius and store it in a variable. Ask user temperature value and store it in int or float type variable.
2. Convert it to Fahrenheit. Using the formula of: f=(c*1.8)+32, convert Celsius to Fahrenheit.
3. Print the final result.
4. Exit.

[30]:
```
celcius = float(input("Enter celcius:"))
f=(celcius*1.8)+32
print(f)
```

```
Enter celcius: 102

215.6
```

Example solutions task 10 CLICK ME

```
celsius=int(input("Enter the temperature in celcius:"))
f=(celsius*1.8)+32
print("Temperature in farenheit is:",f)
```

[ ]: